TEXORA · SESSIONAL NOTES · STUDY SHEET

Straight Knife and Band Knife Cutting Machines

The two powered knives of the cutting room studied part by part: the reciprocating straight knife that travels through the lay, and the stationary band knife against which the work is moved — their parts, working, operation, safety rules and the comparison that assigns each its work.

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 9 · LEVEL 2 TERM 2

1. Aim and place of the study

Aim: to study the construction and working of the straight knife and the band knife cutting machines, operate each under supervision on an exercise lay, and state the safety rules and the proper division of work between them. The cutting process as a whole — requirements, sequence, quality and bundling — is in TE-2207 Topic 6; this study is the two machines themselves.

2. The straight knife machine

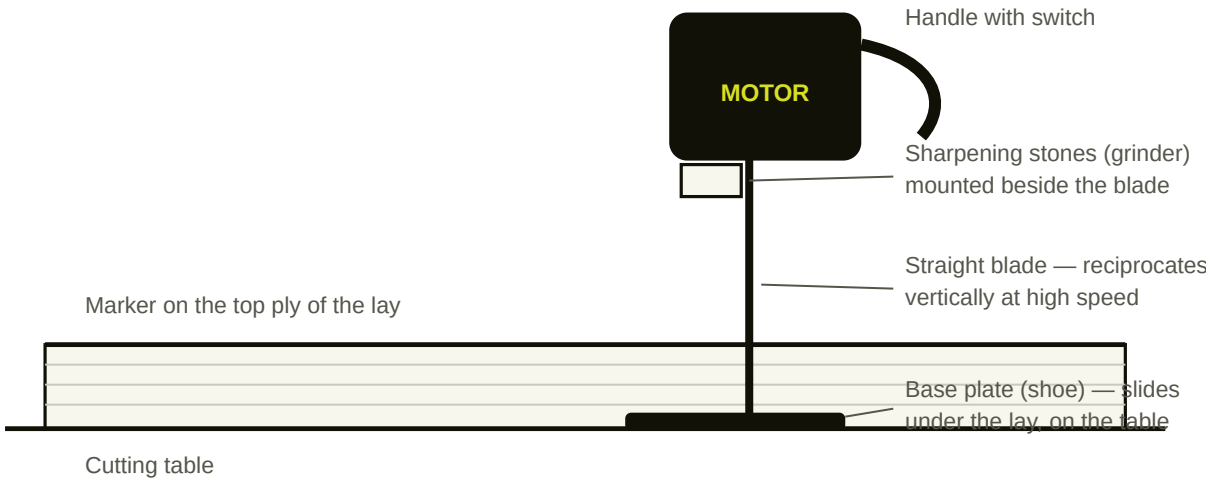


Figure 1 — The straight knife machine at work (side view, schematic — the same machine as TE-2207 Topic 6, Figure 1, here labelled for the parts study).

PART	FUNCTION

Motor and handle	The electric motor on the stand drives the blade and is pushed through the lay by the handle; sizes are rated by blade length (the lay height they can cut) and motor power.
Straight blade	A vertical steel blade, commonly about 13–33 cm long, reciprocating up and down several thousand strokes per minute. Blade edges are plain, wavy (reduces friction and fusing on synthetics) or serrated for special work.
Blade guide and stand	The upright column guides the blade and keeps it vertical — the condition for identical top and bottom plies.
Base plate (shoe)	The flat plate under the stand that slides beneath the bottom ply, supporting the machine as it travels.
Sharpening device	Grinding stones or bands mounted on the machine, pressed against the blade while it runs, so the edge is maintained without removing the blade.
Power cable and trolley	Power reaches the moving machine by an overhead trolley cable, kept off the table and the lay.

Working. The machine travels and the lay stands still: the operator guides the moving blade along the marker line, one hand on the handle, the other holding the lay steady ahead of — never in front of — the blade. It cuts high lays and moderately curved lines, and is the machine that blocks out the lay and cuts the main pieces.

3. The band knife machine

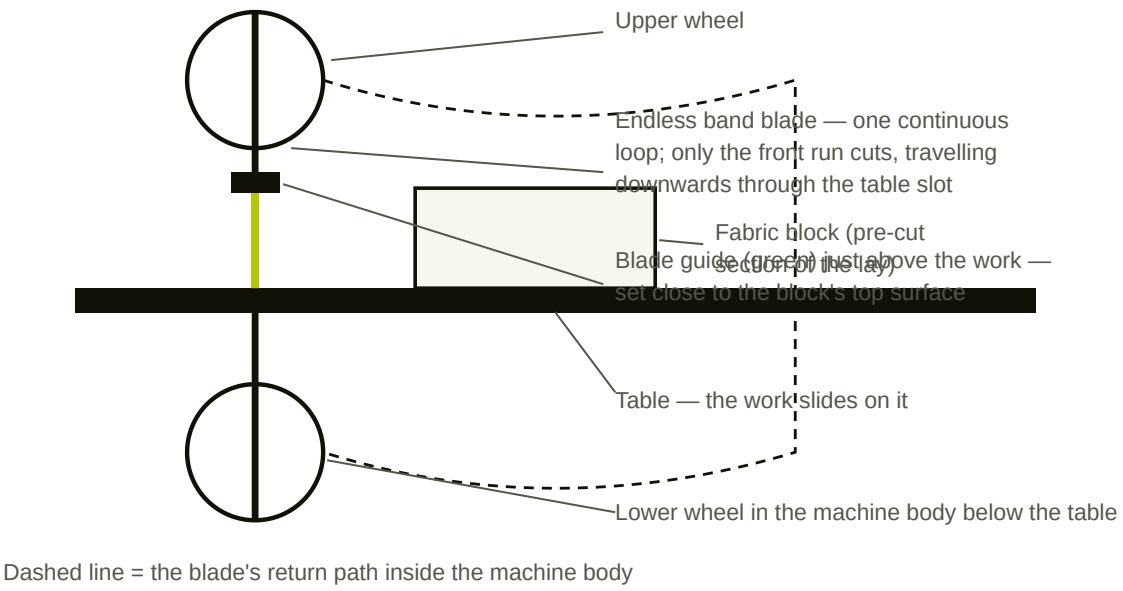


Figure 2 — The band knife machine (front view, schematic): an endless blade loop runs over two wheels; the cutting run is stationary, and the operator moves the fabric block against it.

PART	FUNCTION
Endless blade	A continuous narrow steel band, welded into a loop, with a sharpened front edge; it runs in one direction only — downwards at the cutting position.
Wheels (upper and lower)	Carry and tension the loop; the driven wheel powers it. Blade speed is high and constant, so the cut quality does not depend on the operator's stroke.
Blade guide	A small guide block and rollers just above the work that keep the band from twisting or deflecting sideways under cutting pressure; it is lowered close to the block's surface for every job.
Table	The flat surface on which the fabric block slides; often fitted with an air cushion (air flotation) on production machines so heavy blocks move with fingertip force.
Guard and lamp	The blade above the guide is enclosed by a guard; a lamp concentrates light on the cutting point where both hands are busy.

Working. The machine stands still and the work travels: the operator slides the pre-blocked section of lay on the table and feeds its marked lines into the blade, turning the block to follow curves. Because the blade never leans and its speed never varies, small pieces and sharp curves are cut more accurately here than by any hand-guided machine.

4. Operation steps

Straight knife. (1) Check the marker and lay, clamp the lay ends. (2) Sharpen or change the blade for the fabric; check the blade is vertical. (3) Block out the lay with straight cuts into manageable sections. (4) Precision-cut each piece on its line, curves slowly, keeping the free hand beside or behind the blade line. (5) Notch and drill; strip waste; compare top and bottom plies with the pattern.

Band knife. (1) Receive the blocked sections from the straight-knife work, with the marker lines visible on the top ply. (2) Set the blade guide just above the block's height. (3) Feed the block steadily into the blade, both hands flat on the block, turning the block — not pushing it sideways — to follow the curve. (4) Cut the smallest and most curved pieces last and slowest; collect pieces directly into their sorted piles.

5. Straight knife versus band knife

POINT	STRAIGHT KNIFE	BAND KNIFE
Principle	Blade reciprocates vertically; machine travels through a stationary lay	Endless blade runs continuously; machine stationary, work is moved against the blade
Lay height	High lays — the general production cutter	Lower blocks only — the lay must first be blocked out
Accuracy	Good; depends on the operator's guiding and blade verticality	Higher, especially on small pieces and sharp curves, because the blade is guided and its speed constant
Work assigned	Blocking out; main body pieces; long and moderate lines	Precision cutting of small pieces — collars, cuffs, pockets, curved sections
Sharpening	Built-in stones sharpen the blade on the machine	Blade is sharpened/replaced as a loop; a worn loop is changed
Main risk	Blade leaning or lay dragging gives top-to-bottom size drift	Hands work close to a constantly running blade — discipline and the guard are everything

6. Safety rules (both machines)

- Wear the chain-mail (mesh) glove on the free hand for straight-knife work; keep that hand beside or behind the blade, never ahead of it.
- Clamp the lay before cutting; never hold a tall lay steady by leaning on it near the blade.

- On the band knife, set the blade guide close above the work every time — the exposed blade length is the injury length.
- Never reach across a running blade to move waste; stop the machine or use a stick.
- Power off and blade stopped before changing blades, clearing a jam or cleaning; in the teaching laboratory, blade changes are demonstrated by the instructor first.
- No conversation with an operator whose machine is running; wait for the cut to finish.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the working principle of each machine in one sentence, naming what moves and what stands still.
2. Why must the lay be blocked out before band-knife cutting, and which machine does the blocking?
3. Name the band knife part that prevents the blade twisting on a curve, and state how it is set.
4. A wavy-edged blade is fitted to the straight knife for a synthetic fabric. What fault is it chosen to prevent, and why does it work?
5. Why are collars and cuffs finished on the band knife rather than the straight knife?
6. Give the safety rule that governs the position of the operator's free hand on each machine.
7. Top-ply and bottom-ply sleeves from a straight-knife cutting differ in width. Which machine condition do you check first, and why does it produce exactly this fault?

Answer key

1. Straight knife: a vertically reciprocating blade in a travelling machine cuts a stationary lay. Band knife: a continuously running endless blade stands in one place while the operator moves the fabric block against it.
2. The band knife cuts only low blocks that the operator can slide and turn on the table; a full-height lay is unmanageable and unsafe there. The straight knife does the blocking out, because it travels through the full lay height.
3. The blade guide (guide block and rollers above the work). It is lowered to just above the top surface of the block for every job, minimising the free length of blade that can deflect.
4. Fused edges — plies of synthetic fabric melting together under blade friction heat. The wavy edge reduces the contact and friction between blade and fabric, so less heat is generated as it cuts.
5. They are small pieces with tight curves, where the band knife's guided, constant-speed blade gives the higher accuracy; on the straight knife such pieces would be cut with the least control at the greatest risk of drifting off the line.

6. Straight knife: the free (mesh-gloved) hand holds the lay beside or behind the blade line, never in front of the blade's path. Band knife: both hands stay flat on the block, behind the cutting line, feeding and turning the block — never reaching across or ahead of the blade.
7. Blade verticality. If the blade leans, each lower ply is cut progressively displaced from the line, so plies differ systematically from top to bottom of the lay while every single ply looks cleanly cut.
